

Name: _____ Class: _____

WHITEBRIDGE HIGH SCHOOL



2010

Trial Higher School Certificate

MATHEMATICS EXTENSION 2

Time Allowed: 3 hours
(plus 5 minutes reading time)

Directions to Candidates

- All questions to be completed on writing paper provided
- Commence each question on a new page.
- Marks may be deducted for careless or badly arranged work.

Question 1 (15 marks) Commence each question on a SEPARATE page

a. Evaluate $\int \left(e^x + e^{-\frac{x}{2}} \right)^2 dx.$ **2**

b. Use the substitution $u = 1 + \sin^2 x$ to find $\int \frac{\sin 2x}{\sqrt{1 + \sin^2 x}} dx$ **2**

c. Evaluate in simplest form $\int_0^{\frac{\pi}{4}} \frac{\sec x + \tan x}{\cos x} dx.$ **3**

d. Evaluate in simplest form $\int_0^4 \frac{x - 9}{(x + 1)(x^2 + 9)} dx.$ **4**

e. Use the substitution $t = \tan \frac{x}{2}$ to evaluate, in simplest exact form, **4**

$$\int_0^{\frac{\pi}{2}} \frac{dx}{3 - \cos x - 2 \sin x}.$$

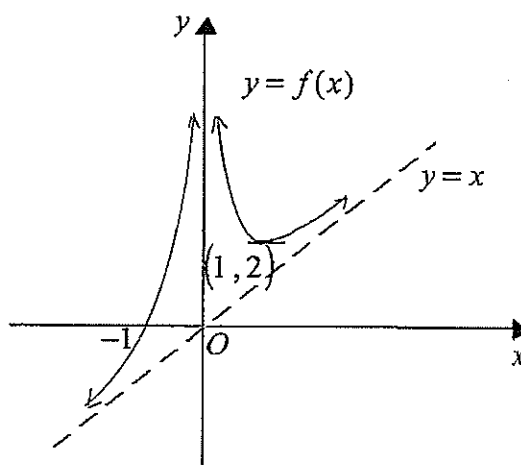
Question 2 (15 marks) Commence each question on a SEPARATE page

- a. If $z = 3 - i$ and $w = 1 + 2i$, find in the form $a + ib$, where a and b are real, the values of
- i. $z - 2w$ **1**
 - ii. $\overline{z w}$ **1**
 - iii. $\frac{z}{w}$ **1**
- b. i. Using the result for $\tan(A - B)$, show that $\tan \frac{\pi}{12} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$. **1**
- ii. Hence, express $z = (\sqrt{3} + 1) + (\sqrt{3} - 1)i$ in modulus argument form. **2**
- iii. Express z^6 in the form $a + ib$, where a and b are real. **1**
- c. i. On an Argand diagram, shade the region where both $|z - 1 - i| \leq \sqrt{2}$ and $0 \leq \arg z \leq \frac{\pi}{4}$. **2**
- ii. Find in simplest exact form the area of the shaded region. **2**
- d. i. If $y = \log_e(\cos \theta + i \sin \theta)$, show that $\frac{dy}{d\theta} = i$. **1**
- ii. Hence, by integration, show that $e^{i\theta} = \cos \theta + i \sin \theta$. **2**
- iii. If $z = e^{i\theta}$, show that $z^4 + \frac{1}{z^4} = 2\cos 4\theta$. **1**

Question 3 (15 marks) Commence each question on a SEPARATE page

- a. The polynomial $P(x) = x^3 - 6x^2 + 9x + c$ has a double zero. **3**
Find any possible values of the real number c .

- b. The graph below shows the curve $y = f(x)$ with asymptotes $x = 0$ and $y = x$.



On separate diagrams, sketch the following graphs showing clearly any intercepts and asymptotes:

- i. $y = |f(x)|$. **1**
 - ii. $y = f(|x|)$. **1**
 - iii. $y = f(x) - x$. **2**
 - iv. $y = \frac{1}{f(x)}$. **2**
- c. $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1$ and the equation $P(x) = 0$ has roots α , β , γ and δ .
- i. Show that the equation $P(x) = 0$ has no integer roots. **1**
 - ii. Show that $P(x) = 0$ has a real root between 0 and 1. **1**
 - iii. Show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = -2$. **2**
 - iv. Hence find the number of real roots of the equation $P(x) = 0$, giving reasons. **2**

Question 4 (15 marks) Commence each question on a SEPARATE page

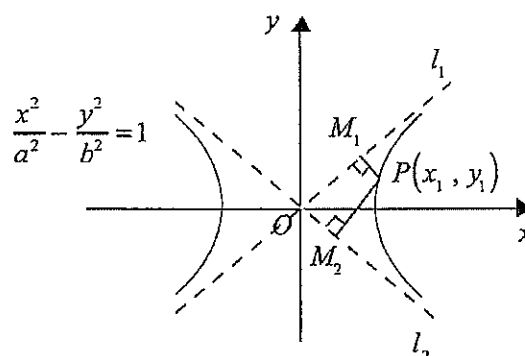
a. For the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$, find

- i. the eccentricity. **1**
- ii. the coordinates of the foci. **1**
- iii. the equations of the directrices. **1**

b. For the curve $y^3 + 2xy + x^2 + 2 = 0$,

- i. show that $\frac{dy}{dx} = \frac{-2(y+x)}{3y^2+2x}$. **2**
- ii. find the coordinates of any stationary points on the curve. **3**

c.



$P(x_1, y_1)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a > b > 0$,

with asymptotes l_1 and l_2 .

M_1 and M_2 are the feet of the perpendiculars from P to l_1 and l_2 respectively.

- i. Show that $PM_1 \times PM_2 = \frac{a^2 b^2}{a^2 + b^2}$. **3**
- ii. Show that $\tan \angle M_1 O M_2 = \frac{2ab}{a^2 - b^2}$. **1**
- iii. Hence find the area of $\triangle PM_1 M_2$ in terms of a and b . **3**

Question 5 (15 marks) Commence each question on a SEPARATE page

a. Let $I_m = \int x^m e^x dx$.

i. Show that $I_m = x^m e^x - m I_{m-1}$. **2**

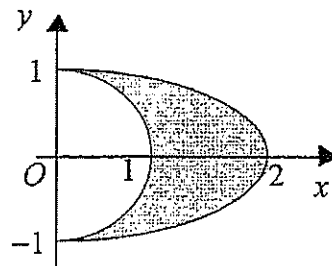
ii. Find the value of $\int_1^2 x^2 e^x dx$. **2**

b. A torus is generated by revolving $x^2 + y^2 \leq 4$ about the line $x = 5$.

i. By using the method of cylindrical shells show that the volume of one shell is given by $\Delta V = 4\pi(5 - x) \sqrt{4 - x^2} \Delta x$. **2**

ii. Hence find the volume of the torus. **2**

c. The base of a solid is the shaded region between the circle $x^2 + y^2 = 1$ and the ellipse $\frac{x^2}{4} + y^2 = 1$ for $x \geq 0$. Vertical cross-sections taken parallel to the x -axis are rectangles with heights equal to the squares of their bases.



i. Show that the volume V of the solid is given by $V = \int_{-1}^1 (1 - y^2)^{\frac{3}{2}} dy$. **2**

ii. It can be shown that $\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$. (Do NOT prove this).

Use the substitution $y = \sin u$ and the result from ii. to find the value of V . **3**

d. Consider the curve defined by the parametric equations $x = t^2 + t - 1$ and $y = te^{2t}$. **2**

Show that $\frac{dy}{dx} = e^{2t}$.

Question 6 (15 marks) Commence each question on a SEPARATE page

a. Solve for x : $\tan^{-1} x + \tan^{-1} (1 - x) = \tan^{-1} \frac{9}{7}$. **4**

b. Consider the function $f(x) = \log_e(1 + \cos x)$, $-2\pi \leq x \leq 2\pi$, where $x \neq \pi$, $x \neq -\pi$.

Show that the function $f(x)$ is even and the curve $y = f(x)$ is concave down for all values of x in the domain. **3**

- c. A particle of mass m kg falls from rest in a medium where the resistance to motion is proportional to the square of its speed and its terminal velocity is 20 ms^{-1} .
The value of g , the acceleration due to gravity is 10 ms^{-1} .
At time t seconds the particle has fallen x metres and acquired a velocity $v \text{ ms}^{-1}$.

i. Explain why $\ddot{x} = \frac{1}{40}(400 - v^2)$. **2**

ii. Find t as a function of v by integration. **2**

iii. Hence, show $\frac{1}{40}v = \frac{\frac{1}{2}\left(e^{\frac{1}{2}t} - e^{-\frac{1}{2}t}\right)}{\left(e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}\right)}$. **2**

iv. Find x as a function of t . **2**

Question 7 (12 marks) Commence each question on a SEPARATE page

- a. The roots of $x^3 - 7x + 6 = 0$ are α , β and γ . **2**
Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

- b. Use mathematical induction to show that, for $n \geq 2$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} \leq 2 - \frac{1}{n} \quad \textbf{4}$$

- c. In a kitchen, the room temperature is 20°C .
Alison makes some coffee, pours a cup and adds milk.
The temperature of the coffee at this point is 80°C .
After p minutes, during which she answers the phone call, the temperature of the coffee has fallen to 35°C .
Then she is delayed for a further 4 minutes by the doorbell, after which the temperature is 27.5°C .
Assuming Newton's law of cooling which states that $T = A + Be^{-kt}$ where T is the temperature in $^\circ\text{C}$, t is the time in minutes, A and B are constants,

- i. find the values of A and B . **2**
- ii. find p , the time that Alison spent on the phone. **3**

Question 8 (7 marks) Commence each question on a SEPARATE page

A car of mass m kg, with speed v metres/second travels around a circular track of radius R metres, inclined at an angle θ to the horizontal and g is the acceleration due to gravity.

- i. Write down the vertical force and horizontal force equations. **1**
- ii. Show that if there is no tendency for the car to slip then $\tan \theta = \frac{v^2}{gR}$. **2**
- iii. Express $\sin \theta$ and $\cos \theta$ in terms of v , g and R . **1**
- iv. If the speed of the car is now halved, prove that the sideways frictional force F on the wheels exerted on the track is given by $F = \frac{3mgv^2}{4\sqrt{v^4 + g^2R^2}}$. **3**

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1:

$$\begin{aligned} \text{a. } \int (e^x + e^{-\frac{x}{2}})^2 dx \\ = \int e^{2x} + 2e^{\frac{x}{2}} + e^{-x} dx \\ = \frac{1}{2}e^{2x} + 4e^{\frac{x}{2}} - e^{-x} + c \end{aligned} \quad 2$$

$$\begin{aligned} \text{b. } u &= 1 + \sin^2 x \\ \frac{du}{dx} &= 2 \sin x \cos x \\ &= \sin 2x \\ \therefore dx &= \frac{du}{\sin 2x} \end{aligned}$$

$$\begin{aligned} \therefore \int \frac{\sin 2x}{\sqrt{1 + \sin^2 x}} dx \\ = \int \frac{\sin 2x}{\sqrt{u}} \cdot \frac{du}{\sin 2x} \\ = \int u^{-\frac{1}{2}} du \\ = 2u^{\frac{1}{2}} + c \\ = 2\sqrt{1 + \sin^2 x} + c \end{aligned} \quad 2$$

$$\begin{aligned} \text{c. } \int_0^{\frac{\pi}{4}} \frac{\sec x + \tan x}{\cos x} dx \\ = \int_0^{\frac{\pi}{4}} \frac{1}{\cos^2 x} + \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} dx \\ = \int_0^{\frac{\pi}{4}} \sec^2 x + \sec x \tan x dx \\ \quad \quad \quad \uparrow \text{stand. int.} \\ = \tan^2 x + \sec x \Big|_0^{\frac{\pi}{4}} \\ = 1 + \sqrt{2} - (0 + 1) \\ = \sqrt{2} \end{aligned} \quad 3$$

$$\begin{aligned} \text{d. } \int_0^4 \frac{x-9}{(x+1)(x^2+9)} dx \\ \therefore \frac{a}{x+1} + \frac{bx+c}{x^2+9} \\ = \frac{a(x^2+9) + (x+1)(bx+c)}{(x+1)(x^2+9)} \\ \therefore x-9 = a(x^2+9) + (x+1)(bx+c) \end{aligned}$$

$$\text{let } x = -1 \therefore -10 = 10a \therefore a = -1$$

$$\begin{aligned} x = 0 \therefore -9 &= 9a + c \\ -9 &= -9 + c \\ c &= 0 \end{aligned}$$

$$\begin{aligned} x = 1 \therefore -8 &= 10a + 2(b+c) \\ -8 &= -10 + 2b \\ 2 &= 2b \\ b &= 1 \end{aligned}$$

$$\begin{aligned} \therefore \int_0^4 \frac{-1}{x+1} + \frac{x}{x^2+9} dx \\ = -\ln(x+1) + \frac{1}{2} \ln(x^2+9) \Big|_0^4 \\ = -\ln 5 + \frac{1}{2} \ln 25 - \left[0 + \frac{1}{2} \ln 9 \right] \\ = -\ln 5 + \ln 5 - \ln 3 \\ = -\ln 3 \end{aligned} \quad 4$$

$$\text{e. } t = \tan \frac{x}{2} \therefore \sin x = \frac{2t}{1+t^2}$$

$$\begin{aligned} \cos x &= \frac{1-t^2}{1+t^2} \\ \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1 + \tan^2 \frac{x}{2}}{2} \\ &= \frac{1+t^2}{2} \end{aligned}$$

$$\begin{aligned} \therefore dx &= \frac{2 dt}{1+t^2} \\ x = \frac{\pi}{2} \therefore t &= 1 \\ x = 0 \therefore t &= 0 \end{aligned}$$

$$\begin{aligned} \therefore \int_0^1 \frac{1}{3 - \frac{1-t^2}{1+t^2} - \frac{4t}{1+t^2}} \cdot \frac{2 dt}{1+t^2} \\ = \int_0^1 \frac{2 dt}{3 + 3t^2 - (1-t^2) - 4t} \\ = \int_0^1 \frac{2 dt}{4t^2 - 4t + 2} \\ = \int_0^1 \frac{dt}{2t^2 - 2t + 1} \\ = \int_0^1 \frac{dt}{1 + 2(t^2 - t + \frac{1}{2}) - \frac{1}{2}} \\ = \int_0^1 \frac{dt}{\frac{1}{2} + 2(t - \frac{1}{2})^2} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^1 \frac{dt}{\frac{1}{4} + (t - \frac{1}{2})^2} \\
 &= \frac{1}{2} \cdot 2 \tan^{-1} \left[\frac{t - \frac{1}{2}}{\frac{1}{2}} \right] \Big|_0^1 \\
 &= \tan^{-1} \left[2(t - \frac{1}{2}) \right] \Big|_0^1 \\
 &= \tan^{-1}(2t - 1) \Big|_0^1 \\
 &= \tan^{-1} 1 - \tan^{-1}(-1) \\
 &= \tan^{-1} 1 + \tan^{-1} 1 \\
 &= 2 \tan^{-1} 1 \\
 &= \frac{\pi}{2}
 \end{aligned}$$

4

Question 2:

$$\begin{aligned}
 \text{a. i. } z - 2w &= 3 - i - 2(1 + 2i) \\
 &= 3 - i - 2 - 4i \\
 &= 1 - 5i
 \end{aligned}$$

1

$$\begin{aligned}
 \text{ii. } z\bar{w} &= (3 - i)(1 - 2i) \\
 &= 3 - 7i - 2 \\
 &= 1 - 7i
 \end{aligned}$$

1

$$\begin{aligned}
 \text{iii. } \frac{z}{w} &= \frac{3 - i}{1 + 2i} \cdot \frac{1 - 2i}{1 - 2i} \\
 &= \frac{3 - 7i - 2}{1 + 4} \\
 &= \frac{1 - 7i}{5}
 \end{aligned}$$

1

$$\text{b. i. } \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\begin{aligned}
 \tan \frac{\pi}{12} &= \tan \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\
 &= \frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{4}}{1 + \tan \frac{\pi}{3} \cdot \tan \frac{\pi}{4}} \\
 &= \frac{\sqrt{3} - 1}{1 + \sqrt{3} \cdot 1} \\
 &= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}
 \end{aligned}$$

1

$$\begin{aligned}
 \text{ii. } |z| &= \sqrt{(\sqrt{3} + 1)^2 + (\sqrt{3} - 1)^2} \\
 &= \sqrt{3 + 2\sqrt{3} + 1 + 3 - 2\sqrt{3} + 1}
 \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{8} \\
 &= 2\sqrt{2}
 \end{aligned}$$

$$\arg z = \tan^{-1} \left[\frac{\sqrt{3} - 1}{\sqrt{3} + 1} \right] = \frac{\pi}{12} \text{ (from ii)}$$

$$\therefore |z| = 2\sqrt{2}, \arg z = \frac{\pi}{12}$$

$$\therefore z = 2\sqrt{2} \operatorname{cis} \frac{\pi}{12}$$

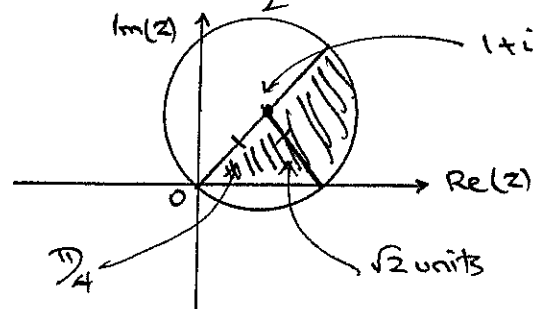
2

$$\begin{aligned}
 \text{iii. } z^6 &= [2\sqrt{2}]^6 \operatorname{cis} \frac{6\pi}{12} \\
 &= 64 \cdot 8 \operatorname{cis} \frac{\pi}{2}
 \end{aligned}$$

$$= 512 \operatorname{cis} \frac{\pi}{2} \rightarrow 0 + 512i$$

1

c.



2

Area = Area of Δ + Area of Sector

$$\begin{aligned}
 &= \frac{1}{2} \times \sqrt{2} \times \sqrt{2} + \frac{1}{4} \cdot \pi \cdot (\sqrt{2})^2 \\
 &= 1 + \frac{\pi}{2}
 \end{aligned}$$

$$\therefore \text{area is } \left(1 + \frac{\pi}{2} \right) u^2$$

2

$$\text{d. i. } y = \log_e (\cos \theta + i \sin \theta)$$

$$\frac{dy}{d\theta} = \frac{-\sin \theta + i \cos \theta}{\cos \theta + i \sin \theta}$$

$$= \frac{i(\cos \theta + i \sin \theta)}{\cos \theta + i \sin \theta} = i$$

1

ii. From i.

$$y = \int i d\theta = i\theta + c$$

But as $\theta = 0$, then

$$\begin{aligned}
 y &= \log_e (\cos 0 + i \sin 0) \\
 &= \log_e 1 \\
 &= 0
 \end{aligned}$$

$$\therefore y = 0, \theta = 0 \therefore c = 0$$

$$\therefore y = i\theta$$

$$\therefore i\theta = \log_e (\cos \theta + i \sin \theta)$$

$$\therefore e^{i\theta} = \cos \theta + i \sin \theta$$

2

iii. $z^4 + z^{-4} = e^{4i\theta} + e^{-4i\theta}$

$$= (\cos 4\theta + i \sin 4\theta) + (\cos(-4\theta) + i \sin(-4\theta))$$

$$= \cos 4\theta + i \sin 4\theta + \cos 4\theta - i \sin 4\theta$$

$$= 2 \cos 4\theta$$

Question 3:

a. $P(x) = x^3 - 6x^2 + 9x + c$

$$P'(x) = 3x^2 - 12x + 9 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3, 1$$

$$\therefore P(3) = 3^3 - 6(3)^2 + 9(3) + c = 0$$

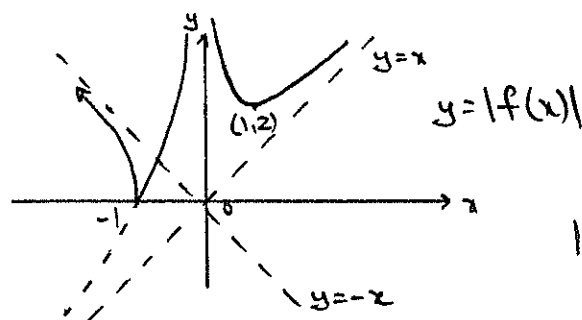
$$\therefore c = 0$$

$$P(1) = 1^3 - 6(1)^2 + 9(1) + c = 0$$

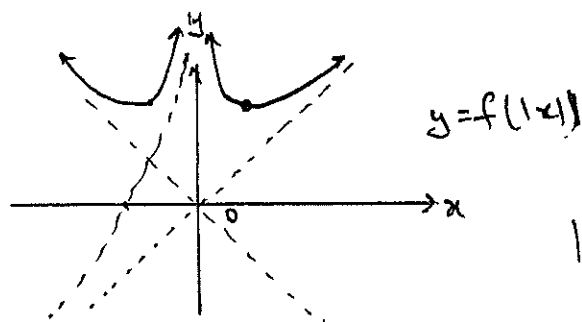
$$c = -4$$

$$\therefore c = 0 \text{ or } -4$$

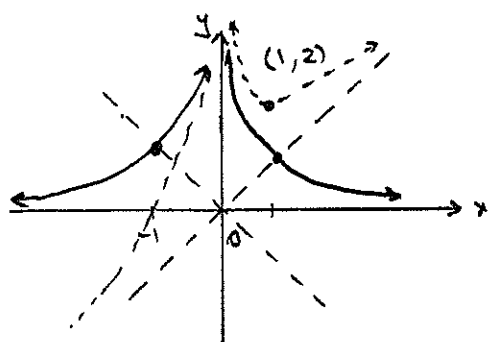
b. i.



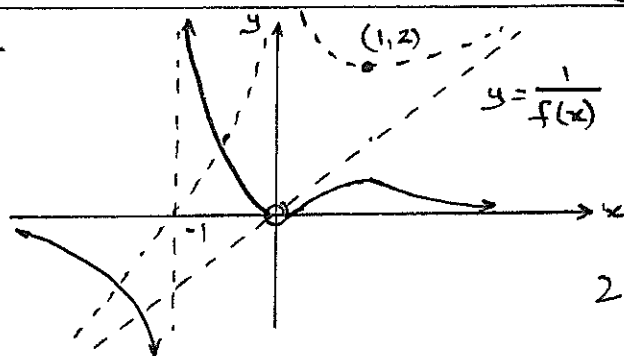
ii.



iii.



iv.



b. $P(x) = x^4 - 2x^3 + 3x^2 - 4x + 1$

i. Only poss. roots are ± 1

$P(1) \neq 0, P(-1) \neq 0 \therefore$ no integer roots

ii. $P(0) = 1$

$$P(1) = 1 - 2 + 3 - 4 + 1 = -1$$

$\therefore$ root lies $0 < x < 1$

iii. As $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

$$= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$$

$$= 2^2 - 2(3)$$

$$= 4 - 6$$

$$= -2$$

iv. As sum of squares of roots < 0 , then at least one root is imaginary.

But as co-eff of $P(x)$ are real, and complex roots are in conjugate pairs, then at least 2 roots are complex. But from ii, real root 2 between 1 & 2 $\therefore$ 2 real roots exist.

Question 4:

a. i. For ellipse, $b^2 = a^2(1 - e^2)$

$$a^2 = 8, b^2 = 4$$

$$\therefore 4 = 8(1 - e^2)$$

$$1 - e^2 = \frac{1}{2}$$

$$e^2 = \frac{1}{2} \therefore e = \frac{1}{\sqrt{2}}$$

ii. foci: $(\pm ae, 0)$

$$= (\pm 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}, 0)$$

$$= (\pm 2, 0)$$

$$\begin{aligned} \text{iii. direct: } x &= \pm \frac{a}{e} \\ &= \pm 2\sqrt{2} \div \frac{1}{\sqrt{2}} \\ \therefore x &= \pm 4 \end{aligned}$$

$$b. y^3 + 2xy + x^2 + 2 = 0 \quad \text{--- ①}$$

$$i. 3y^2 \frac{dy}{dx} + 2\left[y + x \cdot \frac{dy}{dx}\right] + 2x = 0$$

$$3y^2 \frac{dy}{dx} + 2y + 2x \frac{dy}{dx} + 2x = 0$$

$$(3y^2 + 2x) \frac{dy}{dx} = -2y - 2x$$

$$\therefore \frac{dy}{dx} = \frac{-2(x+y)}{3y^2 + 2x} \quad 2$$

$$\begin{aligned} ii. \frac{dy}{dx} &= 0 \therefore -2(x+y) = 0 \\ x+y &= 0 \\ y &= -x \quad \text{--- ②} \end{aligned}$$

$$\text{Subs in ① } -x^3 - 2x^2 + x^2 + 2 = 0$$

$$x^3 + x^2 - 2 = 0$$

$$\text{Let } P(x) = x^3 + x^2 - 2$$

$$P(1) = 0 \therefore x-1 \text{ is factor}$$

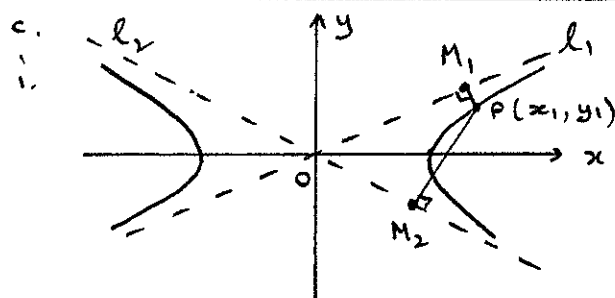
$$\begin{array}{r} x^2 + 2x + 2 \\ x-1 \overline{) x^3 + x^2 + 0x - 2} \\ \underline{x^3 - x^2} \\ 2x^2 + 0x \\ \underline{2x^2 - 2x} \\ 2x - 2 \\ \underline{2x - 2} \\ 0 \end{array}$$

$$\therefore \frac{dy}{dx} = (x-1)(x^2 + 2x + 2)$$

$\nearrow$ No soln as $\Delta < 0$

$$\therefore x=1$$

$$y(1) = -1 \therefore \text{Stat pt } (1, -1) \quad 3$$



$$\text{for hyperbola } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$\text{asymptotes are } y = \pm \frac{b}{a}x$$

$$\text{ie } y = \frac{b}{a}x \text{ and } y = -\frac{b}{a}x$$

$$\therefore l_1: ay = bx \text{ ie } bx - ay = 0$$

$$l_2: ay = -bx \text{ ie } bx + ay = 0$$

Using $\perp$ ar dist. formula:

$$PM_1 \times PM_2 = \left| \frac{bx_1 - ay_1}{\sqrt{b^2 + a^2}} \right| \cdot \left| \frac{bx_1 + ay_1}{\sqrt{b^2 + a^2}} \right|$$

$$= \frac{|b^2x_1^2 - a^2y_1^2|}{b^2 + a^2} \quad \text{--- ①}$$

$$\text{But } \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$$

$$\text{ie } b^2x_1^2 - a^2y_1^2 = a^2b^2 \quad \text{--- ②}$$

Subs ② into ①:

$$PM_1 \times PM_2 = \frac{a^2b^2}{b^2 + a^2} \quad 3$$

$$ii. \text{ using } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\text{and } m_1 = \frac{b}{a}, m_2 = -\frac{b}{a}$$

$$\therefore \tan \theta = \left| \frac{\frac{b}{a} + \frac{b}{a}}{1 + \frac{b}{a} \cdot -\frac{b}{a}} \right|$$

$$= \left| \frac{2b}{a} \div \left(1 - \frac{b^2}{a^2}\right) \right|$$

$$= \left| \frac{2b}{a} \div \left(\frac{a^2 - b^2}{a^2}\right) \right|$$

$$= \left| \frac{2b}{a^2} \times \frac{a^2}{a^2 - b^2} \right|$$

$$\therefore \tan \angle M_1 O M_2 = \frac{2ab}{a^2 - b^2}$$

iii. Now, as $\angle OM_1P = \angle OM_2P = 90^\circ$,

$\therefore OM_1PM_2$ is cyclic quad.

$\therefore \angle M_1OM_2$ and $\angle M_1PM_2$ are supp.

$\therefore \angle M_1PM_2 = 180 - \angle M_1OM_2$

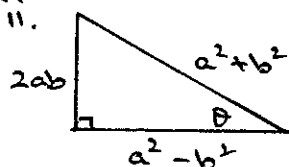
Now, area of $\Delta = \frac{1}{2} ab \sin C$

$$= \frac{1}{2} \cdot PM_1 \times PM_2 \times \sin(180^\circ - \angle M_1OM_2)$$

$$= \frac{1}{2} \cdot \frac{a^2b^2}{a^2+b^2} \cdot \sin \angle M_1OM_2$$

(as $\sin(180-A) = \sin A$)

From ii.



$$\therefore \sin \theta = \frac{2ab}{a^2+b^2}$$

$$\therefore \text{Area} = \frac{1}{2} \cdot \frac{a^2b^2}{a^2+b^2} \cdot \frac{2ab}{a^2+b^2}$$

$$= \frac{a^3b^3}{(a^2+b^2)^2} u^2$$

3

Question 5:

a. i. $I_m = \int x^m e^x dx$

let $u = x^m$

$v' = e^x$

$u' = mx^{m-1}$

$v = e^x$

$$\therefore I_m = x^m e^x - \int mx^{m-1} \cdot e^x dx$$

$$= x^m e^x - m I_{m-1}$$

2

ii. $\int_1^2 x^2 e^x dx$

$$= x^2 e^x \Big|_1^2 - 2 I_1$$

$$= 4e^2 - e - 2 \left[\int_1^2 x e^x dx \right]$$

$$= 4e^2 - e - 2 \left[x e^x \Big|_1^2 - I_0 \right]$$

$$= 4e^2 - e - 2(2e^2 - e) + 2 \int_1^2 x dx$$

$$= 4e^2 - e - 4e^2 + 2e + 2 \int_1^2 e^x dx$$

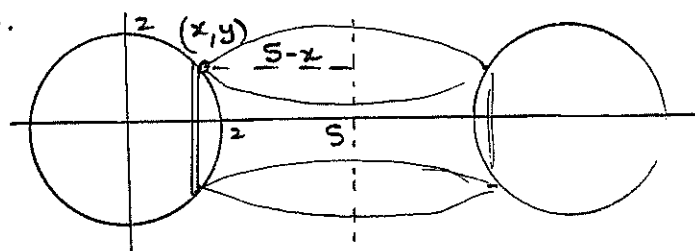
$$= 4e^2 - e - 4e^2 + 2e + 2(e^2 - e)$$

$$= e + 2e^2 - 2e$$

$$= 2e^2 - e$$

2

b.



$$x^2 + y^2 = 4$$

$$\therefore y = \pm \sqrt{4-x^2}$$

radius of $5-x$

$$\Delta V = 2\pi (\text{radius})(\text{height})$$

$$= 2\pi (5-x) \cdot 2\sqrt{4-x^2} \cdot \Delta x$$

$$\therefore \Delta V = 4\pi (5-x) \sqrt{4-x^2} \cdot \Delta x$$

ii. $\int_{-2}^2 4\pi (5-x) \sqrt{4-x^2} dx$

$$= 4\pi \int_{-2}^2 5\sqrt{4-x^2} - x\sqrt{4-x^2} dx$$

semi-circle

odd function

$$\therefore \int = 0$$

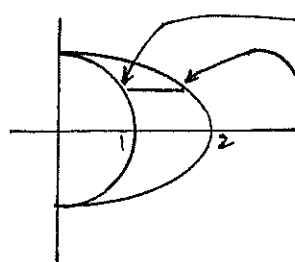
$$= 20\pi \cdot \frac{1}{2} \cdot \pi \cdot 2^2$$

$$= 40\pi^2$$

$$\therefore \text{volume is } 40\pi^2 u^3$$

c.

i.



$$x^2 = 1 - y^2$$

$$\therefore x = \sqrt{1-y^2}$$

$$\frac{x^2}{1} = 1 - y^2$$

$$\therefore \frac{x}{2} = \sqrt{1-y^2}$$

$$x = 2\sqrt{1-y^2}$$

$$\therefore \text{length} : 2\sqrt{1-y^2} - \sqrt{1-y^2}$$

$$= \sqrt{1-y^2}$$

$$\text{Also, height} = (\sqrt{1-y^2})^2$$

$$= 1-y^2$$

$\therefore$ Area of rectangle

$$= (1-y^2)^{\frac{3}{2}} \cdot (1-y^2)$$

$$= (1-y^2)^{\frac{5}{2}}$$

$$\therefore V = \int_{-1}^1 (1-y^2)^{\frac{3}{2}} dy$$

2

ii. let $y = \sin u$

$$\therefore \frac{dy}{du} = \cos u \therefore dy = \cos u du$$

$$\begin{aligned}
 v &= 2 \int_0^1 (1-y^2)^{\frac{3}{2}} dy \quad (\text{as } f' \text{ is even}) \\
 &= 2 \int_0^{\frac{\pi}{2}} (1-\sin^2 u)^{\frac{3}{2}} \cdot \cos u \cdot du \\
 &= 2 \int_0^{\frac{\pi}{2}} \cos^3 u \cdot \cos u \, du \\
 &= 2 \int_0^{\frac{\pi}{2}} \cos^4 u \, du \\
 &= \frac{2}{8} \int_0^{\frac{\pi}{2}} \cos 4u + 4 \cos 2u + 3 \, du \\
 &= \frac{1}{4} \left[\frac{1}{4} \sin 4u + 2 \sin 2u + 3u \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{4} \left[\frac{1}{4} \sin 2\pi + 2 \sin \pi + 3\frac{\pi}{2} - (0) \right] \\
 &= \frac{3\pi}{8}
 \end{aligned}$$

$$d. \frac{dy}{dt} = 2t+1$$

$$\begin{aligned}
 \frac{dy}{dt} &= 1 \cdot e^{2t} + t \cdot 2e^{2t} \\
 &= e^{2t}(1+2t)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{dy}{dt} \cdot \frac{dt}{dx} \\
 &= e^{2t}(1+2t) \cdot \frac{1}{1+2t} \\
 &= e^{2t}
 \end{aligned}$$

Question 6:

$$a. \text{ Let } \alpha = \tan^{-1} x \quad \therefore \tan \alpha = x$$

$$\beta = \tan^{-1}(1-x) \quad \therefore \tan \beta = 1-x$$

$$\therefore \alpha + \beta = \tan^{-1} \frac{9}{7}$$

$$\therefore \tan(\alpha + \beta) = \frac{9}{7}$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{9}{7}$$

$$\frac{x + 1 - x}{1 - x(1-x)} = \frac{9}{7}$$

$$7 = 9[1 - x + x^2]$$

$$7 = 9 - 9x + 9x^2$$

$$9x^2 - 9x + 2 = 0$$

$$(3x-1)(3x-2) = 0$$

$$x = \frac{1}{3}, \frac{2}{3}$$

$$\begin{aligned}
 b. i. f(-x) &= \log_e(1 + \cos(-x)) \\
 &= \log_e(1 + \cos x) = f(x)
 \end{aligned}$$

 $\therefore$ even

$$f'(x) = \frac{-\sin x}{1 + \cos x}$$

$$f''(x) = \frac{(1 + \cos x) \cdot -\cos x + \sin x \cdot -\sin x}{(1 + \cos x)^2}$$

$$= \frac{-\cos x - \cos^2 x - \sin^2 x}{(1 + \cos x)^2}$$

$$= \frac{-\cos x - (\sin^2 x + \cos^2 x)}{(1 + \cos x)^2}$$

$$= \frac{-(1 + \cos x)}{(1 + \cos x)^2}$$

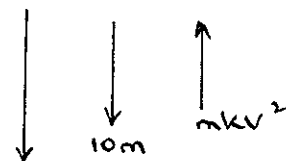
$$= \frac{-1}{1 + \cos x}$$

But $-1 \leq \cos x \leq 1$, and as

$$x \neq \pi, x \neq -\pi \quad \therefore \cos x \neq -1$$

$$\therefore f''(x) < 0 \text{ for all } x \text{ in domain.}$$

c.



$$m\ddot{x} = 10m - mkv^2$$

$$\ddot{x} = 10 - kv^2$$

Terminal velocity is $\ddot{x} = 0$

$$\therefore 10 - kv^2 = 0$$

$$\text{But } v = 20 \quad \therefore 10 - 400k = 0$$

$$k = \frac{1}{40}$$

$$\therefore \ddot{x} = 10 - \frac{v^2}{40}$$

$$= \frac{1}{40}(400 - v^2)$$

$$ii. \frac{dv}{dt} = \frac{1}{40}(400 - v^2)$$

$$\frac{dt}{dv} = \frac{40}{400 - v^2}$$

$$= \frac{40}{(20-v)(20+v)}$$

$$\therefore \frac{a}{20-v} + \frac{b}{20+v} = \frac{a(20+v)+b(20-v)}{(20-v)(20+v)}$$

$$\therefore a(20+v)+b(20-v)=40$$

$$\text{Let } v=20 \therefore 40a=40 \therefore a=1$$

$$v=-20 \therefore 40b=40 \therefore b=1$$

$$\therefore \frac{dt}{dv} = \frac{1}{20-v} + \frac{1}{20+v}$$

$$\therefore t = -\ln(20-v) + \ln(20+v) + c$$

$$\therefore t = \ln\left(\frac{20+v}{20-v}\right) + c \quad 2$$

$$\text{Now, } t=0, v=0 \therefore c=0$$

$$\therefore t = \ln\left(\frac{20+v}{20-v}\right)$$

$$\text{iii. } \therefore e^t = \frac{20+v}{20-v}$$

$$(20-v)e^t = 20+v$$

$$20e^t - ve^t = 20+v$$

$$20e^t - 20 = v(1+e^t)$$

$$\therefore 20(e^t - 1) = v(e^t + 1)$$

$$20e^{\frac{1}{2}t}(e^{\frac{1}{2}t} - e^{-\frac{1}{2}t}) = v e^{\frac{1}{2}t}(e^{\frac{1}{2}t} + e^{-\frac{1}{2}t})$$

$$\therefore v = \frac{20(e^{\frac{1}{2}t} - e^{-\frac{1}{2}t})}{e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}}$$

$$\therefore \frac{1}{40}v = \frac{\frac{1}{2}(e^{\frac{1}{2}t} - e^{-\frac{1}{2}t})}{e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}} \quad 2$$

$$\text{iv. } \frac{1}{40} \frac{dx}{dt} = \frac{\frac{1}{2}(e^{\frac{1}{2}t} - e^{-\frac{1}{2}t})}{e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}}$$

$$\frac{1}{40} \cdot x = \ln(e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}) + c$$

$$t=0, x=0$$

$$\therefore 0 = \ln(e^0 + e^0) + c$$

$$\therefore c = -\frac{1}{2}\ln 2$$

$$\therefore \frac{1}{40}x = \ln(e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}) - \frac{1}{2}\ln 2$$

$$\frac{1}{40}x = \ln\left[\frac{e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}}{2}\right]$$

$$\therefore x = 40 \ln\left[\frac{e^{\frac{1}{2}t} + e^{-\frac{1}{2}t}}{2}\right] \quad 2$$

Question 7:

a. If α is root,

$$\text{then } \alpha^3 - 7\alpha + 6 = 0$$

$$\text{ie } \alpha^3 = 7\alpha - 6 \quad \text{--- ①}$$

$$\text{Similarly, } \beta^3 = 7\beta - 6 \quad \text{--- ②}$$

$$\gamma^3 = 7\gamma - 6 \quad \text{--- ③}$$

$$\text{①} + \text{②} + \text{③}:$$

$$\alpha^3 + \beta^3 + \gamma^3 = 7\alpha - 6 + 7\beta - 6 + 7\gamma - 6$$

$$= 7(\alpha + \beta + \gamma) - 18$$

$$= 7(0) - 18 \quad 2$$

$$= -18 \quad \text{as } \alpha + \beta + \gamma = 0$$

b. Step 1: Prove true for $n=2$

$$\therefore \text{LHS} = \frac{1}{1^2} + \frac{1}{2^2}$$

$$= 1 + \frac{1}{4}$$

$$= 1\frac{1}{4}$$

$$\text{RHS} = 2 - \frac{1}{2}$$

$$= 1\frac{1}{2}$$

$$\text{as } 1\frac{1}{4} \leq 1\frac{1}{2} \therefore \text{LHS} \leq \text{RHS}$$

$$\therefore \text{true for } n=2$$

Step 2: Assume true for $n=k$

$$\therefore \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{k^2} \leq 2 - \frac{1}{k}$$

Now, prove true for $n=k+1$

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1}$$

$$\text{LHS} = \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2}$$

$$\leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2}$$

$$= 2 - \left[\frac{1}{k} - \frac{1}{(k+1)^2} \right]$$

$$= 2 - \left[\frac{(k+1)^2 - 1}{k(k+1)^2} \right]$$

$$= 2 - \left[\frac{k^2 + 2k + 1 - 1}{k(k+1)^2} \right]$$

$$= 2 - \frac{1}{k+1} \left[\frac{k^2+2k}{k(k+1)} \right]$$

$$= 2 - \frac{1}{k+1} \left[\frac{k(k+2)}{k(k+1)} \right]$$

$$\leq 2 - \frac{1}{k+1}, \text{ as } \frac{k+2}{k+1} > 1$$

$\therefore$ true for $n=k+1$

Step 3: As true for $n=1$, then

true for $n=2, 3, \dots$ for $n \geq 2$

b. i. $T = A + Be^{-kt}$

When $t = -\infty$, $T = 20$

$$\therefore 20 = A + B \cancel{e^{-k(-\infty)}} \therefore A = 20$$

$$\therefore T = 20 + Be^{-kt}$$

When $t = 0$, $T = 80$

$$\therefore 80 = 20 + B \cancel{e^{-k(0)}} \therefore B = 60 \quad 2$$

$$\therefore T = 20 + 60e^{-kt} \therefore A = 20, B = 60$$

ii. When $t = p$, $T = 35$

$$35 = 20 + 60e^{-kp}$$

$$60e^{-kp} = 15$$

$$e^{-kp} = 0.25 \quad \text{--- (1)}$$

Now, $t = p+4$, $T = 27.5$

$$\therefore 27.5 = 20 + 60e^{-k(p+4)}$$

$$60e^{-k(p+4)} = 7.5$$

$$\therefore e^{-kp-4k} = 0.125$$

$$\therefore \frac{e^{-kp}}{e^{4k}} = 0.125 \quad \text{--- (2)}$$

$$\therefore \frac{0.25}{e^{4k}} = 0.125$$

$$\therefore e^{4k} = 2$$

$$4k = \ln 2$$

$$k = \frac{\ln 2}{4}$$

Subs in (1) $e^{-p\left(\frac{\ln 2}{4}\right)} = 0.25$

$$-p\left(\frac{\ln 2}{4}\right) = \ln 0.25$$

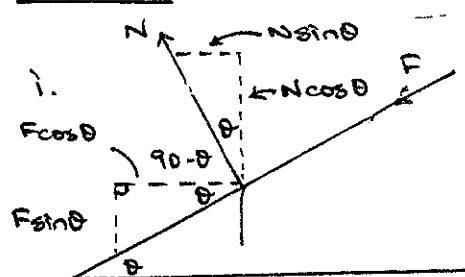
$$p = -\ln 0.25 \div \left(\frac{\ln 2}{4}\right)$$

$$= 8$$

3

$\therefore$ Alison spent 8 min on phone.

Question 8:



Vertical: $N \cos \theta - F \sin \theta - mg = 0 \quad \text{--- (3)}$

Horiz: $N \sin \theta + F \cos \theta = \frac{mv^2}{R} \quad \text{--- (4)}$

ii. If no slip, then $F = 0$

$$\therefore N \cos \theta = mg \quad \text{--- (1)}$$

$$N \sin \theta = \frac{mv^2}{R} \quad \text{--- (2)}$$

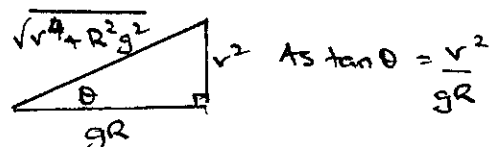
$$\therefore (2) \div (1) \quad \tan \theta = \frac{mv^2}{R} \div mg$$

$$= \frac{v^2}{R} \times \frac{1}{g}$$

$$= \frac{v^2}{Rg}$$

2

iii.



$$\therefore \sin \theta = \frac{v^2}{\sqrt{v^4 + R^2 g^2}} \quad \cos \theta = \frac{Rg}{\sqrt{v^4 + R^2 g^2}} \quad \text{--- (5)}$$

iv.

Now, mult. (3) by $\sin \theta$:

$$N \sin \theta \cos \theta - F \sin^2 \theta - mg \sin \theta = 0 \quad \text{--- (5)}$$

and mult (4) by $\cos \theta$:

$$N \sin \theta \cos \theta + F \cos^2 \theta - \frac{mv^2}{R} \cos \theta = 0 \quad \text{--- (6)}$$

$$\textcircled{5} - \textcircled{5} F(\sin^2\theta + \cos^2\theta) + mg \sin\theta - \frac{mv^2}{R} \cos\theta = 0$$

But vel is halved $\therefore$ subs $\frac{v}{2}$

$$\therefore F = \frac{mv^2}{4R} \cos\theta - mg \sin\theta = 0$$

Now, using $\textcircled{*}$

$$F = \frac{mv^2}{4R} \cdot \frac{gR}{\sqrt{v^4 + R^2g^2}} - mg \cdot \frac{v^2}{\sqrt{v^4 + R^2g^2}}$$

$$\therefore F = \left| \frac{mgv^2}{4\sqrt{v^4 + R^2g^2}} - \frac{mgv^2}{\sqrt{v^4 + R^2g^2}} \right|$$

$$= \frac{3mv^2g}{4\sqrt{v^4 + g^2R^2}}$$

Note: F must be opp. direction